National University of Computer & Emerging Sciences, Karachi
FAST SCHOOL OF COMPUTING
Fall 2025 - Computer Organization & Assembly Language (EE 2003)

Assignment 02 | Points: 100

Deadline: November 01, 2025

Q -1 : What will be the final values in EAX, EBX, **and** the stack **after the following instructions execute?**

```
mov eax, 3
mov ebx, 7
push eax
push ebx
pop eax
push 10
pop ebx
pop eax
```

Q-2: Implement the following pseudocode in x86 assembly language. All variables are 32-bit signed integers:

```
if (num1 > num2 && num3 != num4)
   result = num1 + num3;
else if (num2 == num4)
   result = num2 - num1;
else
   result = 0;
```

Q-3 : Implement the following pseudocode in x86 assembly language. All variables are 32-bit signed integers.

```
sum = 0; i = 0;
while (i < count)
{
          if (array[i] > 0)
      sum = sum + array[i];
          else
      sum = sum - 1;
          i = i + 1;
}
```

Q-4 : Consider the following code:

```
MOV AX, 0H
MOV ECX, 0AH
DOLOOP:
DEC AX
LOOP DOLOOP
```

What is the value of the AX register after completion of the DOLOOP?

Q-5 : In the following code sequence, show the value of AL (in hexadecimal format only) after each shift or rotate instruction has executed:

```
MOV AL,0D4H
SHL AL,3 ;
MOV AL,0D4H
SAL AL,3 ;
STC
MOV AL,0D4H
ROL AL,1 ;
STC
MOV AL,0D4H
RCR AL,3 ;
```

Q-6 : Convert the following High Level Language code to equivalent assembly language:

```
  While (op1 <= op2)
  {
if (op1 >= x && op1 =< y)
{
z += 10;
   }
else
{
 z -= 10;
 }
    op1--;
  }
```

Q-7: Write an assembly language program that checks whether a signed byte value in register AL is positive, negative, or zero, and displays a message accordingly (use Irvine library procedures).

```
INCLUDE Irvine32.inc
```

Q-8: Write an assembly program that multiplies two 16-bit signed integers stored in variables num1 and num2, stores the 32-bit result in result, and displays it using WriteInt.

```
INCLUDE Irvine32.inc
```

Q-9: Implement the following C++ code in assembly language, using the flag registers(ZF and CF). Assume that all variables are 16-bit Un-signed integers:

```
int array[] = {10,60,20,33,72,89,45,65,72,18};
int sample = 50;
int ArraySize = sizeof array / sizeof sample;
int index = 0; int sum = 0;
while( index < ArraySize )
{
  if( array[index] <= sample )
          { sum += array[index]; }
 index++; }
```

Q-10 : In Assembly language, multiplication by constants can be optimized using shifts and additions, since any multiplier can be expressed as a sum of powers of 2.

$36 = 2^5 + 2^2 = 32 + 4$ $\qquad$ $EAX \times 36 = (EAX \times 32) + (EAX \times 4)$

Using this concept:

(a) Write an equivalent Assembly code sequence (without using the MUL instruction) to multiply the unsigned integer value in EAX by 26 using shift and addition instructions.

(b) If EAX initially contains 2, show the final value in EAX after executing your code.

---------------------------------- Best of Luck ----------------------------------